

MPI - Collective Communication and Computation

- involves all processes in communicator
 - MPI_COMM_WORLD is default
 - can create own subsets
 - MPI-2+ can create even bigger sets if dynamic process allocation is supported
- programs using only collective communication are easy to understand
 - every process does roughly the same thing
 - no inventive communication patterns
- functions for collective communication are optimized
 - devised by experts
 - detailed implementation depends on infrastructure
 - existing protocols in network infrastructure (broadcast)
- all collective functions must be called by all processes in the communicator
- functions work with any number of processes from 1 onwards
- all collective functions are blocking (MPI-1, MPI-2)
- there are no tags
- basic data types (MPI-1, MPI-2)
- types of collectives
 - synchronization
 - data transfer
 - collective computation

Synchronization

- MPI_Barrier
 - rarely used
 - for performance measurements

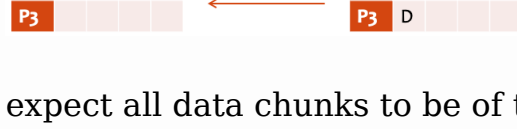
Data Transfer

- MPI_Bcast
 - one to all (broadcast)

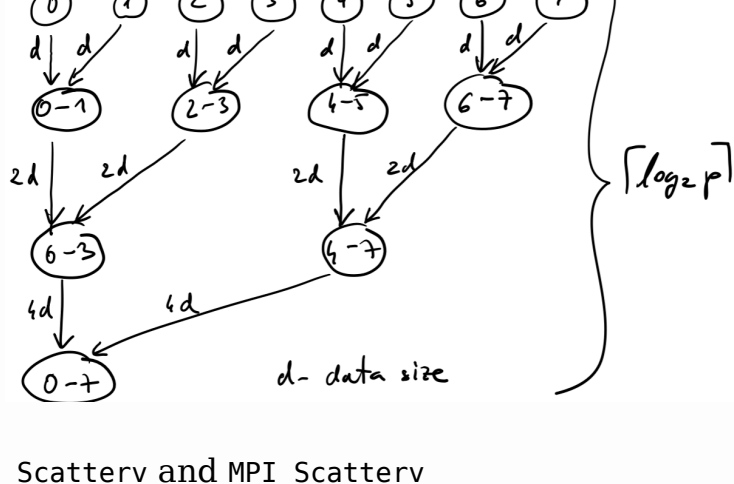


- MPI_Scatter and MPI_Gather

- scatters or gathers data across processes in the same communicator



- expect all data chunks to be of the same size
- root process takes care of one data chunk
- some parameters are valid on side of sender, some on side of receiver
- tree-like implementation of gather



- MPI_Scatterv and MPI_Gatherv
 - more general but slower functions
 - size of data chunk can vary
- MPI_Allgather
 - combines gather and broadcast
 - can be efficiently implemented by only one pass of the tree



- MPI_Alltoall

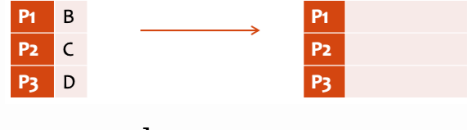
- transpose of data
- tricky to implement efficiently



- example: The Conway's Game of Life
 - the board is split to horizontal stripes
 - cells on borders are exchanged via MPI_Sendrecv
 - code:
 - [conway.c](#)
 - [board.h](#)
 - [conway.sh](#)

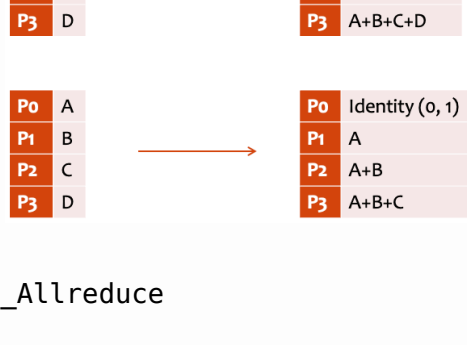
Collective computation

- MPI_Reduce
 - reduces data from several processes
 - reduce operations
 - extreme: MPI_MIN, MPI_MAX
 - sum and product: MPI_SUM, MPI_PROD
 - logical operations: MPI_LAND, MPI_LOR, MPI_LXOR
 - bit-wise operations: MPI_BAND, MPI BOR, MPI_BXOR
 - extreme with location: MPI_MAXLOC, MPI_MINLOC



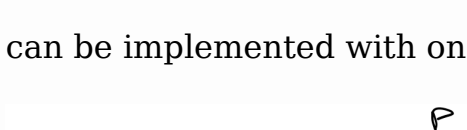
- MPI_Scan and MPI_Exscan

- inclusive and exclusive scan

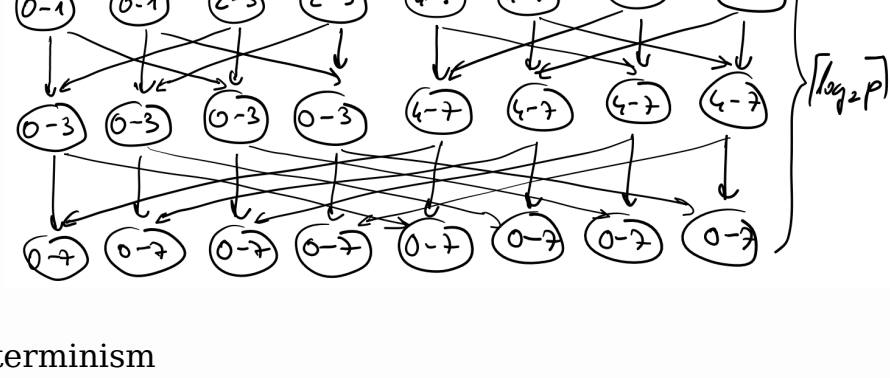


- MPI_Allreduce

- combination of reduce and broadcast



- can be implemented with one pass of a tree



- determinism
 - rounding error, truncation, depends on order of computation
 - MPI does not guarantee the same result on the same input
 - encouraged but not required
 - not all applications need it
 - more efficient implementations of collectives are possible without it
- example: reduce with location information
 - combined data type
 - code:
 - [maxloc.c](#)
 - [maxloc.sh](#)

Advanced MPI Features

- data types
- communicators
- virtual topology
 - reflects actual system configuration
 - Cartesian, graph
- MPI-IO
- collective functions
 - neighborhood
 - immediate